

50 g Lab Cup · BX-01

BEST

NEW

Closed-cell pour is **50.00 g B (includes 5.00 g HFO-1233zd(E)) + 41.40 g A**. HFO is mandatory. Cream/gel are sped up. Mix 4–5 s. Weigh to 0.01 g.

Cup size	
B-side (HFO in)	50.00 g
of which HFO-1233zd(E)	5.00 g
A-side (PMDI 31 % NCO)	41.40 g
Total	91.40 g
Index	125
Mix	A into B · 4–5 s · 2 500–3 000 rpm
B cup temperature	15–18 °C after HFO

Trial header	
Date	
Operator	
Drum · high-OH	TB-307 / PE 104
Drum · flow	TEROL 250 / PE 103
Cup T	A 21–23 °C · B 15–18 °C
Cup ID	BX-01-50-

BX-01 weigh sheet · 50.00 g B · HFO mandatory

Weigh in this order. Tare a dry, closable B cup. Water before catalysts. **HFO last**, cup at 15–18 °C, cap at once. Write the actual in the last column.

#	INGREDIENT	TARGET G	ACTUAL G
1	ISOEXTER TB-307 / KIMpol PE 104	20.65	
2	TEROL 250 / KIMpol PE 103	12.40	
3	CREAPOL PD 230	2.00	
4	TCPP	3.00	
5	TEP	2.00	
6	Reactive FR	1.00	
7	Emulsifier	0.875	

#	INGREDIENT	TARGET G	ACTUAL G
8	Closed-cell silicone	0.75	
9	Water (deionized / distilled)	0.60	
10	PC-15	0.725	
11	DABCO 2040	0.550	
12	CAT-T	0.275	
13	PC-37	0.175	
14	HFO-1233zd(E) — last, cold, cap	5.00	
	B-side total	50.000	
A	PMDI, 31.0 % NCO	41.40	

HFO 5.00 g is 10 pbw on 50 g. Water 0.60 g is 1.20 pbw. Do not pad a short cup with extra water. If HFO flashes off while weighing, recap and reweigh — do not add extra water to compensate.

Checklist · before the pour NEW

Safety and bench

- ☐ Gloves, goggles, lab coat
- ☐ Fume hood or good ventilation
- ☐ Balance reads 0.01 g and is level
- ☐ Two dry cups, labeled A and B
- ☐ Timer ready (cream / gel / rise / tack-free)
- ☐ Thermometer · A at 21–23 °C · B at 15–18 °C after HFO
- ☐ Mixer 2 500–3 000 rpm, clean blade
- ☐ Paper towel / waste cup for the blade

Drums and grades

- ☐ High-OH drum is TB-307 or PE 104 — write which
- ☐ Flow drum is TEROL 250 or PE 103 — write which
- ☐ PD 230, TCPP, TEP, reactive FR on the bench
- ☐ Closed-cell silicone — not the open-cell grade
- ☐ PC-15, PC-37, CAT-T, DABCO 2040
- ☐ PMDI is 31.0 % NCO (write lot / %NCO)
- ☐ HFO-1233zd(E) on the bench — mandatory for this cup
- ☐ HFO cylinder / bottle cold · hood on · no open flame
- ☐ Closable B cup ready (HFO boils at ~19 °C)
- ☐ No sucrose, Mannich, K-15, DMCHA or pentane in this cup

Checklist · pour and record

Pour

- ☐ Resin mixed homogeneous, then chilled to 15–18 °C
- ☐ HFO 5.00 g weighed last · cup capped at once
- ☐ A weighed 41.40 g in a dry cup
- ☐ Pour A into B — start the clock at mix start
- ☐ Mix 4–5 s only — cream is 3–6 s, then do not touch
- ☐ Leave cup 24 h before cutting

Record (write)

Cream	s	target 3–6
Gel	s	target 8–15
Rise	s	
Tack-free	s	target 15–28
Collapse / shrink / scorch		
Free-rise density	kg/m³	target 32–38
Pass?		3 cups, no collapse, times repeat

Half-cup (25 g B, 2.50 g HFO, 20.70 g A) is allowed only if the 0.175 g PC-37 weigh is trusted. Prefer 50.00 g B. Mix 4–5 s. Open-cell 50 g is the next sheet.

50 g Lab Cup · BX-02 Open Cell

BESTNEW

Open-cell 50 g cup. **No HFO** on this pour. Only if sucrose is on the floor. Do not blend with BX-01 or OC-01.

Cup size	
B-side	50.00 g
A-side (PMDI 31 % NCO)	65.70 g at index 105
Total	115.70 g
Index	100–110 (calc. 105)
Mix	A into B · 4–5 s · 2 500–3 000 rpm
Cup temperature	A and B 21–23 °C

Trial header	
Date	
Operator	
Sucrose grade / OH#	assumed OH 400 for A-side
Flow polyester	TEROL 250 / PE 103
Cup T (A / B)	21–23 °C
Cup ID	BX-02-50-

BX-02 weigh sheet · 50.00 g B

Weigh in this order. Tare a dry B cup. Water before catalysts. Open-cell silicone only. No HFO. A-side assumes sucrose OH 400 — re-index when the lot OH is known.

#	INGREDIENT	TARGET G	ACTUAL G
1	Sucrose / sucrose polyol	19.00	
2	TEROL 250 / KIMpol PE 103	8.00	
3	TEP	10.45	
4	Reactive FR	3.50	
5	Ecosurf SA-9	2.00	
6	Open-cell silicone	0.75	
7	Water (deionized / distilled)	2.60	
8	PC-37	0.85	

#	INGREDIENT	TARGET G	ACTUAL G
9	CAT-T	0.80	
10	PC-15	1.40	
11	BDMAEE or DEDEE	0.45	
12	Biocide	0.20	
	B-side total	50.00	
A	PMDI, 31.0 % NCO	65.70	

Water 2.60 g is 5.2 pbw. A-side 65.70 g is index 105 at sucrose OH 400. If sucrose OH is 350, A ≈ 62 g. If 450, A ≈ 70 g. Do not force 1:1. No HFO on this cup.

Checklist · open-cell 50 g NEW

Before

- ☐ Gloves, goggles, lab coat
- ☐ Balance 0.01 g · two dry cups A and B
- ☐ A and B at 21–23 °C
- ☐ Sucrose polyol on the floor — write grade / OH#
- ☐ TEROL 250 or PE 103 — write which
- ☐ Open-cell silicone — not the closed-cell grade
- ☐ PC-37, CAT-T, PC-15, BDMAEE, biocide
- ☐ No HFO / pentane on this cup

Pour and record

- ☐ B mixed homogeneous
- ☐ A weighed 65.70 g (or re-indexed)
- ☐ Pour A into B · mix 4–5 s · clock from mix start
- ☐ Do not touch after cream · cut at 24 h

Cream

s

target 2–4

Rise

s

Tack-free

s

Free-rise density

kg/m³

target 16–22

Cells

open / tight / collapse

OC-01 memo screen · 50.00 g B

High-water System C only. Not the best card. A-side 20–28 g PMDI — water 12.00 g dominates. No HFO.

INGREDIENT	G	INGREDIENT	G
Sucrose / sucrose polyol	16.50	Reactive FR	4.00

INGREDIENT	G	INGREDIENT	G
TEP	8.50	Ecosurf SA-9	3.00
Open-cell silicone	0.60	Water	12.00
PC-37	1.20	CAT-T	1.20
PC-15	2.10	BDMAEE / biocide	0.50 / 0.40
B-side	50.00	A-side PMDI	20–28

If cells stay closed on BX-02: +0.20 g water or –0.10 g PC-15. If it collapses: +0.15 g PC-15. Do not pour BX-02 and OC-01 as one blend.